

ESAT PRACTICE TEST SERIES

ESAT 2024-25 Recall Mock

Mathematics 1 • Mathematics 2 • Chemistry

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard, recall-informed original practice with full worked solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 1 answer key

Q1	C	Q2	A	Q3	D	Q4	E	Q5	D	Q6	B	Q7	B	Q8	C	Q9	A
Q10	E	Q11	C	Q12	B	Q13	E	Q14	E	Q15	A	Q16	D	Q17	A	Q18	B
Q19	D	Q20	A	Q21	D	Q22	C	Q23	E	Q24	B	Q25	D	Q26	E	Q27	C

Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

M1-01

Ratio and replacement

Answer **C**

Question 1

A tank contains an alloy in which copper and zinc are in the mass ratio **5 : 3**. Exactly **8 kg** of the well-mixed alloy is removed and replaced by **8 kg** of pure zinc. The new copper-to-zinc ratio is **3 : 2**.

What was the original mass of alloy in the tank?

A 80 kg**B** 192 kg**C** 200 kg**D** $\frac{200}{9}$ kg**E** 320 kg

WORKED SOLUTION

Let the original mass be M kg. Removing 8 kg of the well-mixed alloy leaves

$$\text{copper} = \frac{5}{8}(M - 8), \quad \text{zinc} = \frac{3}{8}(M - 8).$$

After 8 kg of zinc is added,

$$\frac{\frac{5}{8}(M - 8)}{\frac{3}{8}(M - 8) + 8} = \frac{3}{2}.$$

Hence $10(M - 8) = 9(M - 8) + 192$, so $M = 200$.

Therefore the correct option is **C**.

Trap check: Option B ignores the removed alloy before the zinc is added; option A treats the removal as changing the copper-to-zinc ratio even though a well-mixed sample preserves it; option D reverses the final ratio; option E treats the removed mass as pure copper.

Question 2

For positive integers a and b , the highest common factor of a and b is 6 , and their lowest common multiple is 180 .

How many *ordered* pairs (a, b) satisfy these conditions?

A 8**B** 4**C** 6**D** 3**E** 16**WORKED SOLUTION**

Write $a = 6x$ and $b = 6y$, where x and y are coprime. Then

$$6xy = 180 \implies xy = 30 = 2 \cdot 3 \cdot 5.$$

Each prime factor must be assigned wholly to either x or y . There are $2^3 = 8$ assignments, and order matters.

Therefore the correct option is **A**.

Trap check: Option B counts unordered pairs, C omits the assignments with one reduced factor equal to 1, D merely counts the distinct primes, and E incorrectly includes negative ordered pairs despite the positivity condition.

Question 3

A vehicle travels a route of length D . On one day it covers the first third of the route at speed v and the remainder at speed $2v$, taking **48** minutes in total.

On another day it covers the first half at speed $4v/3$, stops for **6** minutes, and then covers the second half at constant speed kv . The total time is again **48** minutes.

What is k ?

A 2

B $\frac{9}{4}$

C $\frac{5}{2}$

D $\frac{12}{5}$

E $\frac{8}{3}$

WORKED SOLUTION

The first journey gives

$$\frac{D/3}{v} + \frac{2D/3}{2v} = \frac{2D}{3v} = 48,$$

so $D/v = 72$ minutes. On the second day the first half takes

$$\frac{D/2}{4v/3} = \frac{3}{8} \frac{D}{v} = 27 \text{ minutes.}$$

After the six-minute stop, **15** minutes remain. Therefore

$$\frac{D/2}{kv} = \frac{36}{k} = 15, \quad k = \frac{12}{5}.$$

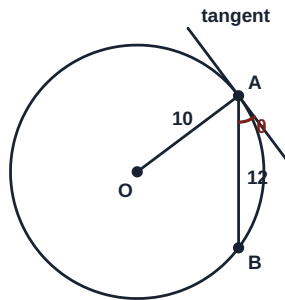
Therefore the correct option is **D**.

Trap check: Do not average segment speeds without checking the associated times; here the arithmetic mean happens to agree because the two segment times are equal.

Question 4

A circle has centre O , radius 10 , and chord AB of length 12 . A tangent is drawn at A . Let θ be the acute angle between the tangent and chord AB .

What is $\tan \theta$?



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{5}{4}$

E $\frac{3}{4}$

WORKED SOLUTION

Let M be the midpoint of AB . The perpendicular from the centre to a chord bisects it, so $AM = 6$. Hence

$$OM = \sqrt{10^2 - 6^2} = 8.$$

If $\alpha = \angle OAB$, then $\tan \alpha = 8/6 = 4/3$. Since the tangent is perpendicular to OA , $\theta = 90^\circ - \alpha$, and therefore

$$\tan \theta = \frac{1}{\tan \alpha} = \frac{3}{4}.$$

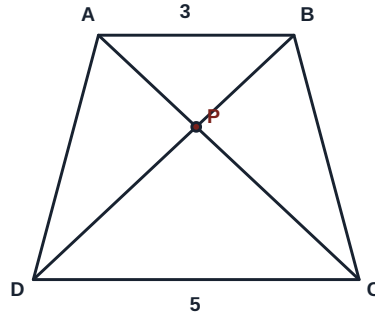
Therefore the correct option is E.

Trap check: The value $4/3$ is $\tan \angle OAB$, not the tangent–chord angle requested.

Question 5

In trapezium $ABCD$, $AB \parallel CD$. The diagonals AC and BD meet at P , and $AB : CD = 3 : 5$. The area of triangle APB is 27.

What is the area of triangle CPD ?



A 45

B 125

C $\frac{243}{25}$ **D 75**

E 192

WORKED SOLUTION

Triangles APB and CPD are similar: they have equal vertically opposite angles at P , and corresponding angles from $AB \parallel CD$. Their linear scale factor is $3 : 5$, so their areas are in the ratio $9 : 25$.

$$27 : \text{area}(CPD) = 9 : 25.$$

Therefore $\text{area}(CPD) = 27 \times 25/9 = 75$.

Therefore the correct option is D.

Trap check: Option A uses the linear scale factor, B cubes it, C reverses the area factor, and E gives the area of the whole trapezium rather than triangle CPD .

Question 6

The sequence begins

$$7, 14, 25, 40, \dots$$

and has constant second differences. Let n be the smallest index for which the n th term exceeds 500, and let e be the amount by which that term exceeds 500.

Which pair (n, e) is correct?

A (15, 31)

B (16, 32)

C (16, 31)

D (15, 32)

E (17, 69)

WORKED SOLUTION

The first differences are 7, 11, 15, ..., so the second difference is 4. Fitting $u_n = 2n^2 + bn + c$ gives

$$u_n = 2n^2 + n + 4.$$

Now $u_{15} = 469$, whereas $u_{16} = 532$. Thus $n = 16$, and the excess is $e = 532 - 500 = 32$.

Therefore the correct option is B.

Trap check: A and D stop one term too early; C subtracts incorrectly after finding the correct index; E moves to the next term after the threshold has already been crossed.

Question 7

A sample has recorded mass **2.400 kg**, correct to the nearest **0.010 kg**, and recorded volume **800 cm³**, correct to the nearest **10 cm³**.

If its density is $\rho \text{ g cm}^{-3}$, which statement gives the exact bounds for ρ ?

A $\frac{479}{159} \leq \rho < \frac{481}{161}$

B $\frac{479}{161} < \rho < \frac{481}{159}$

C $\frac{480}{161} \leq \rho < \frac{480}{159}$

D $\frac{2395}{800} \leq \rho < \frac{2405}{800}$

E $2.995 \leq \rho < 3.005$

WORKED SOLUTION

In grams and cubic centimetres,

$$2395 \leq m < 2405, \quad 795 \leq V < 805.$$

For positive quantities, m/V is smallest as m takes its lower bound and V approaches its excluded upper bound; it is greatest as m approaches its excluded upper bound and V takes its lower bound. Hence

$$\frac{2395}{805} < \rho < \frac{2405}{795}.$$

Dividing numerator and denominator by 5 gives

$$\frac{479}{161} < \rho < \frac{481}{159}.$$

Therefore the correct option is B.

Trap check: The lower endpoint is also excluded: the recorded-volume interval has $V < 805$, so density can approach $2395/805$ but cannot equal it.

Question 8

A bag contains **4** red, **3** blue and **2** green counters. Two counters are drawn at random without replacement.

Given that the counters are different colours, what is the probability that one of them is green?

A $\frac{5}{13}$

B $\frac{6}{13}$

C $\frac{7}{13}$

D $\frac{7}{18}$

E $\frac{14}{27}$

WORKED SOLUTION

Count unordered pairs. The number of different-colour pairs is

$$4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2 = 26.$$

Of these, the pairs involving green number $4 \cdot 2 + 3 \cdot 2 = 14$. Hence the conditional probability is

$$\frac{14}{26} = \frac{7}{13}.$$

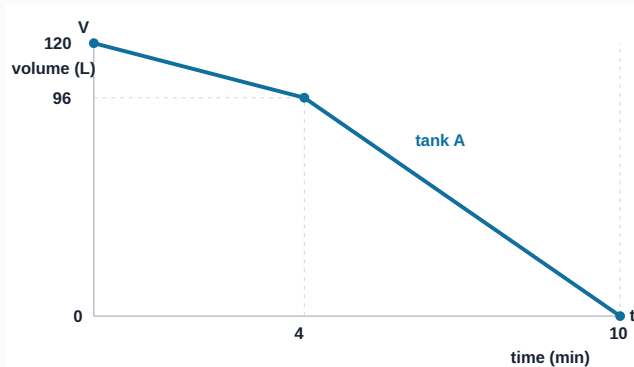
Therefore the correct option is C.

Trap check: The denominator must contain only different-colour pairs, not all $\binom{9}{2}$ pairs.

Question 9

The graph shows the volume V litres of water remaining in tank A, t minutes after draining begins. At $t = 0$, an empty tank B starts filling at a constant rate of **12** litres per minute.

At what time do the two tanks contain the same volume?

**A** $\frac{40}{7}$ min**B** 5 min**C** 6 min**D** $\frac{20}{3}$ min**E** 8 min**WORKED SOLUTION**

At $t = 4$, tank A contains **96** L. From $t = 4$ to $t = 10$, its volume falls by **96** L in **6** min, so the rate is **16** L/min. On this segment,

$$V_A = 96 - 16(t - 4) = 160 - 16t.$$

Tank B contains $V_B = 12t$. Equating volumes gives

$$160 - 16t = 12t \implies t = \frac{40}{7}.$$

This lies between **4** and **10**, so the correct segment was used.

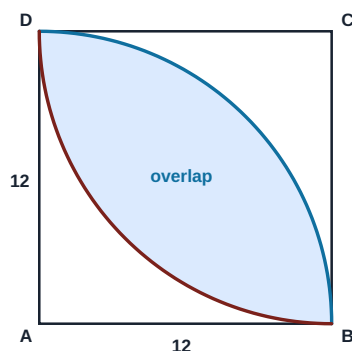
Therefore the correct option is A.

Trap check: The first segment's rule gives a time outside its interval; the change of gradient at 4 minutes matters.

Question 10

A square $ABCD$ has side length **12**. Inside it, quarter-circles of radius **12** are drawn with centres **A** and **C**. The shaded lens is the region lying inside both quarter-circles.

What is the exact area of the shaded region?



A $36\pi - 72$

B 72π

C $144 - 36\pi$

D $72\pi - 72$

E $72\pi - 144$

WORKED SOLUTION

Each arc BD subtends 90° at its centre because adjacent sides of the square are perpendicular. One half of the lens is a 90° sector of radius **12**, less a right triangle with legs **12**:

$$\frac{1}{4}\pi(12)^2 - \frac{1}{2}(12)(12) = 36\pi - 72.$$

The lens has two congruent halves, so its area is

$$2(36\pi - 72) = 72\pi - 144.$$

Therefore the correct option is **E**.

Trap check: A uses only half the lens, B subtracts no triangles, C takes an incorrect square complement, and D subtracts only one of the two right triangles.

Question 11

Let $u = \sqrt{3} + \sqrt{2}$ and $v = \sqrt{3} - \sqrt{2}$.

What is the value of

$$\frac{u^6 + v^6}{u^2 + v^2} ?$$

A 10

B 94

C 97

D 100

E 103

F 970

WORKED SOLUTION

Since $uv = 1$ and $u + v = 2\sqrt{3}$,

$$u^2 + v^2 = (u + v)^2 - 2uv = 10.$$

Using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ with $a = u^2$, $b = v^2$,

$$u^6 + v^6 = 10^3 - 3(uv)^2(10) = 1000 - 30 = 970.$$

The required quotient is $970/10 = 97$.

Therefore the correct option is **C**.

Trap check: Option A stops at $u^2 + v^2$, B incorrectly uses $(uv)^2 = 2$, D omits the cubic cross-term, and E changes its sign.

Question 12

The point P lies on the line $y = 2x + 1$ and is equidistant from $A = (0, 4)$ and $B = (6, 0)$. The origin is O .

What is OP^2 ?

A 169

B 218

C 196

D 224

E 49

WORKED SOLUTION

Write $P = (x, 2x + 1)$. Equating squared distances avoids square roots:

$$x^2 + (2x - 3)^2 = (x - 6)^2 + (2x + 1)^2.$$

The quadratic terms cancel, leaving $-12x + 9 = -8x + 37$. Hence $-4x = 28$, so $x = -7$. Therefore $P = (-7, -13)$, and

$$OP^2 = (-7)^2 + (-13)^2 = 49 + 169 = 218.$$

Therefore the correct option is B.

Trap check: A uses only the square of the y -coordinate, C omits x^2 , D introduces a cross-term as if calculating $(x + y)^2$, and E reports only x^2 .

Question 13

Two distinct cards are selected at random from cards numbered **1** to **8**. It is known that their product is even and their sum is **not** divisible by **3**.

What is the probability that their product is divisible by **4**?

A $\frac{9}{22}$

B $\frac{1}{2}$

C $\frac{5}{14}$

D $\frac{7}{11}$

E $\frac{9}{14}$

WORKED SOLUTION

Use unordered pairs. There are **22** pairs with an even product. Of the **10** pairs whose sum is divisible by **3**, two have odd product, so eight are excluded from those **22**. The conditioned sample space therefore has $22 - 8 = 14$ pairs.

The nine admissible pairs with product divisible by **4** are

$$(2, 6), (2, 8), (1, 4), (3, 4), (4, 6), (4, 7), (6, 8), (3, 8), (5, 8).$$

The required probability is **9/14**.

Therefore the correct option is E.

Trap check: A uses the 22 even-product pairs as the denominator and ignores the sum condition; B assumes an unsupported half-and-half split; C counts only a restricted subset centred on the cards 4 and 6; D counts products divisible by 4 before enforcing the sum condition.

Question 14

Let

$$N = 2^4 \cdot 3^3 \cdot 5^2.$$

How many positive divisors of N are divisible by exactly one of **12** and **18**?**A** 9**B** 6**C** 18**D** 33**E** 15**F** 51**WORKED SOLUTION**Divisors divisible by $12 = 2^2 \cdot 3$ number

$$(3)(3)(3) = 27.$$

Divisors divisible by $18 = 2 \cdot 3^2$ number

$$(4)(2)(3) = 24.$$

Those divisible by both are divisible by $\text{lcm}(12, 18) = 36 = 2^2 3^2$, giving

$$(3)(2)(3) = 18.$$

To count divisors in exactly one set, subtract the intersection twice:

$$27 + 24 - 2(18) = 15.$$

Therefore the correct option is E.**Trap check:** Options A and B count divisors in only one named set separately, C counts the intersection, and D is the ordinary inclusion–exclusion total for at least one condition.

Question 15

Containers **A**, **B**, **C** initially hold liquid in the ratio **1 : 3 : 5**. Half of the liquid in **A** is transferred to **B**. Then half of the resulting amount in **B** is transferred to **C**. Finally, **C** contains **100** litres more than **A**.

How much liquid was there originally in total?

A 144 L**B** 200 L**C** 150 L**D** 180 L**E** 108 L**WORKED SOLUTION**

Let the initial amounts be **k**, **3k**, **5k**. After the first transfer,

$$A = \frac{1}{2}k, \quad B = \frac{7}{2}k.$$

Half of this new amount in **B** is $\frac{7k}{4}$, so

$$B = \frac{7}{4}k, \quad C = 5k + \frac{7}{4}k = \frac{27}{4}k.$$

Thus $C - A = (27/4 - 1/2)k = 25k/4 = 100$, giving $k = 16$. The original total was $9k = 144$ L.

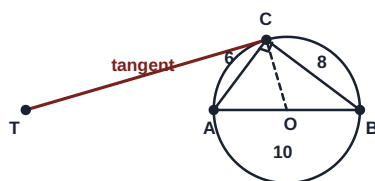
Therefore the correct option is **A**.

Trap check: B ignores the second transfer, C transfers half of B's initial amount, D compares the final amounts in C and B, and E reports C's final amount rather than the original total.

Question 16

Triangle ABC is right-angled at C , with $AC = 6$, $BC = 8$ and $AB = 10$. The circle with diameter AB is shown. The tangent to the circle at C meets the line AB , extended beyond A , at T .

What is CT ?



A $\frac{60}{7}$

B 12

C 15

D $\frac{120}{7}$

E 20

WORKED SOLUTION

Place $A = (0, 0)$ and $B = (10, 0)$. The 6-8-10 triangle gives $C = (18/5, 24/5)$, and the centre is $O = (5, 0)$. The gradient of OC is

$$\frac{24/5}{18/5 - 5} = -\frac{24}{7}.$$

The tangent is perpendicular to OC , so its gradient is $7/24$. Its equation through C is

$$y - \frac{24}{5} = \frac{7}{24} \left(x - \frac{18}{5} \right).$$

Setting $y = 0$ gives $T = (-90/7, 0)$. Therefore

$$CT = \sqrt{\left(\frac{18}{5} + \frac{90}{7} \right)^2 + \left(\frac{24}{5} \right)^2} = \frac{120}{7}.$$

Therefore the correct option is **D**.

Trap check: The tangent is perpendicular to the radius OC , not to either side of triangle ABC .

Question 17

A right-angled triangle has perpendicular sides **9** and **12**. A square is placed with one side on the hypotenuse and the opposite side parallel to it, with the other two vertices lying on the perpendicular sides.

What is the side length of the square?

A $\frac{180}{37}$

B $\frac{36}{7}$

C $\frac{60}{11}$

D $\frac{72}{13}$

E 6

WORKED SOLUTION

The hypotenuse has length **15**, and the altitude from the right angle to the hypotenuse is

$$h = \frac{2 \times (\frac{1}{2})(9)(12)}{15} = \frac{36}{5}.$$

If the square side is s , the small triangle above it is similar to the original triangle. Its scale factor is $(h - s)/h$, so its base, which is a side of the square, satisfies

$$s = 15 \frac{h - s}{h}.$$

Thus $s(h + 15) = 15h$, and with $h = 36/5$,

$$s = \frac{15(36/5)}{15 + 36/5} = \frac{180}{37}.$$

Therefore the correct option is **A**.

Trap check: This is not the standard square placed at the right-angle vertex; here a side of the square lies on the hypotenuse.

Question 18

A data set contains only the values shown.

Value	1	2	3	4	5
Frequency	2	a	4	b	2

There are **20** observations and their mean is **3.1**. Two observations are selected at random without replacement.

What is the probability that they have equal values?

A $\frac{39}{200}$

B $\frac{39}{190}$

C $\frac{31}{190}$

D $\frac{39}{380}$

E $\frac{3}{10}$

WORKED SOLUTION

The total frequency gives $a + b = 12$. The total of all observations is $20(3.1) = 62$, so

$$2 + 2a + 12 + 4b + 10 = 62 \implies a + 2b = 19.$$

Hence $b = 7$ and $a = 5$. The number of unordered equal-value pairs is

$$\binom{2}{2} + \binom{5}{2} + \binom{4}{2} + \binom{7}{2} + \binom{2}{2} = 1 + 10 + 6 + 21 + 1 = 39.$$

There are $\binom{20}{2} = 190$ unordered pairs in total, so the probability is $39/190$.

Therefore the correct option is B.

Trap check: Using 20^2 as the denominator models selection with replacement and also permits selecting the same observation twice.

Question 19

A price is increased by $p\%$, where $p > 0$. The increased price is then reduced by $(p + 10)\%$. The final price is **12%** below the original price.

What is p ?

A -20

B 12

C 22

D 10

E 20

WORKED SOLUTION

Let $x = p/100$. The two multipliers give

$$(1 + x)(0.9 - x) = 0.88.$$

Expanding and simplifying,

$$0.9 - 0.1x - x^2 = 0.88 \implies x^2 + 0.1x - 0.02 = 0.$$

Hence $(x - 0.1)(x + 0.2) = 0$. Since $p > 0$, $x = 0.1$, so $p = 10$.

Therefore the correct option is **D**.

Trap check: A is the rejected algebraic root, B copies the final percentage, C adds 12 and 10, and E neglects the product term in the two successive multipliers.

Question 20

What is the smallest positive integer n for which both $72n$ is a perfect square and $50n$ is a perfect cube?

A 20000**B** 5000**C** 10000**D** 16000**E** 40000**WORKED SOLUTION**

Write $n = 2^a 3^b 5^c$ with the least possible non-negative exponents. Since $72 = 2^3 3^2$, the exponents in $72n$ must be even. Since $50 = 2 \cdot 5^2$, the exponents in $50n$ must be multiples of 3. The least simultaneous choices are

$$a = 5, \quad b = 0, \quad c = 4.$$

Thus $n = 2^5 \cdot 5^4 = 32 \cdot 625 = 20000$.

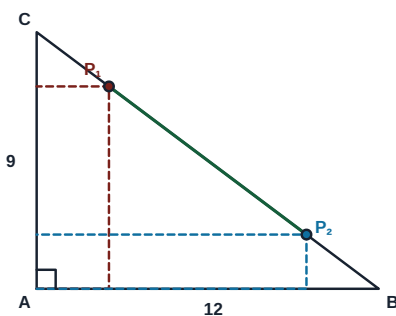
Therefore the correct option is **A**.

Trap check: The same n must satisfy both exponent conditions simultaneously; satisfying only the square condition is insufficient.

Question 21

Triangle ABC is right-angled at A , with $AB = 12$ and $AC = 9$. For a point P on the hypotenuse, perpendiculars from P to AB and AC form a rectangle having A and P as opposite vertices.

There are two positions, P_1 and P_2 , for which the rectangle has area 18. What is P_1P_2 ?



A $3\sqrt{3}$

B $4\sqrt{3}$

C 5

D $5\sqrt{3}$

E $7\sqrt{3}$

WORKED SOLUTION

Place $A = (0, 0)$, $B = (12, 0)$ and $C = (0, 9)$. If $P = (x, y)$, the hypotenuse equation is $y = 9 - \frac{3x}{4}$. The rectangle area condition gives

$$x \left(9 - \frac{3}{4}x \right) = 18 \implies x^2 - 12x + 24 = 0.$$

The two roots are $6 \pm 2\sqrt{3}$, so the horizontal separation of P_1, P_2 is $4\sqrt{3}$. Their vertical separation is $(\frac{3}{4})(4\sqrt{3}) = 3\sqrt{3}$. Hence

$$P_1P_2 = \sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2} = 5\sqrt{3}.$$

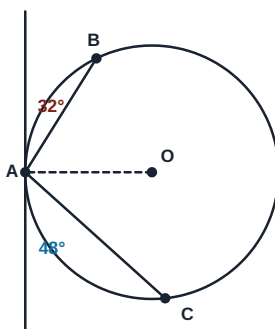
Therefore the correct option is **D**.

Trap check: The horizontal root difference is only $4\sqrt{3}$; P_1P_2 also includes the vertical separation.

Question 22

At a point **A** on a circle, a tangent is drawn. Chords **AB** and **AC** lie on opposite sides of the radius through **A**. The acute angles between the tangent and the chords **AB** and **AC** are 32° and 48° , respectively.

What is the smaller angle **BOC**, where **O** is the centre?



A 100°

B 80°

C 160°

D 200°

E 40°

WORKED SOLUTION

By the tangent–chord theorem,

$$\angle ACB = 32^\circ, \quad \angle ABC = 48^\circ.$$

Therefore $\angle BAC = 180^\circ - 32^\circ - 48^\circ = 100^\circ$. The arc **BC** not containing **A** has central angle 200° . The smaller angle at the centre is therefore

$$360^\circ - 200^\circ = 160^\circ.$$

Therefore the correct option is **C**.

Trap check: A gives the inscribed angle, B adds the two tangent–chord angles without doubling, D selects the reflex central angle, and E doubles their difference.

Question 23

An increasing arithmetic sequence has positive integer terms. Its terms satisfy

$$u_6^2 = u_3 u_{11} \quad \text{and} \quad u_{11} - u_3 = 48.$$

What is the first term of the sequence?

A 6

B 9

C 12

D 18

E 15

WORKED SOLUTION

Let the first term be a and common difference be $d > 0$. Then

$$u_3 = a + 2d, \quad u_6 = a + 5d, \quad u_{11} = a + 10d.$$

The difference condition gives $8d = 48$, so $d = 6$. Expanding the other condition,

$$(a + 5d)^2 = (a + 2d)(a + 10d)$$

$$a^2 + 10ad + 25d^2 = a^2 + 12ad + 20d^2,$$

so $5d^2 = 2ad$. Since $d > 0$, $a = 5d/2 = 15$.

Therefore the correct option is **E**.

Trap check: The identity is multiplicative, not the ordinary arithmetic-mean relation for u_3 , u_6 and u_{11} .

Question 24

Two fair six-sided dice are thrown.

Given that the product of the two scores is divisible by 6, what is the probability that their sum is prime?

A $\frac{7}{15}$

B $\frac{8}{15}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{9}$

WORKED SOLUTION

The 15 ordered outcomes whose product is divisible by 6 are

$(1, 6); (2, 3), (2, 6); (3, 2), (3, 4), (3, 6); (4, 3), (4, 6); (5, 6); (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6).$

Of these, the eight outcomes

$(1, 6), (6, 1), (2, 3), (3, 2), (3, 4), (4, 3), (5, 6), (6, 5)$

have prime sum. The required probability is therefore $\frac{8}{15}$.

Therefore the correct option is B.

Trap check: A misses one favourable ordered outcome, C counts unordered pairs, D includes one non-prime sum, and E uses the eight favourable outcomes over the original 36-outcome sample space.

Question 25

The arithmetic sequence

$$5, 12, 19, 26, \dots$$

contains terms that are divisible by **11** but not divisible by **3**. What is the position of the **15th** such term?

A 214

B 225

C 236

D 247

E 258

F 159

WORKED SOLUTION

The n th term is $7n - 2$. Divisibility by **11** requires

$$7n \equiv 2 \pmod{11},$$

so $n \equiv 5 \pmod{11}$. Write $n = 5 + 11k$, where $k \geq 0$. The term is divisible by **3** precisely when $n \equiv 2 \pmod{3}$. Since

$$5 + 11k \equiv 2 + 2k \pmod{3},$$

this happens when $2k \equiv 0 \pmod{3}$, so $k \equiv 0 \pmod{3}$. These values must be excluded. The allowed values of k are therefore

$$1, 2, 4, 5, 7, 8, \dots$$

There are two allowed values in each block of three. The **15th** allowed value is $k = 22$, giving

$$n = 5 + 11(22) = 247.$$

Indeed, $7(247) - 2 = 1727 = 11 \times 157$, which is not divisible by **3**.

Therefore the correct option is D.

Trap check: C excludes the wrong residue class of k ; F counts the 15th multiple-of-11 candidate without any exclusion; A, B and E are one-block or one-candidate counting shifts.

Question 26

A positive real number x satisfies

$$x + \frac{1}{x} = 3.$$

For positive integers n , define

$$A_n = x^n + \frac{1}{x^n}.$$

What is the remainder when A_{2025} is divided by 10?

A 2

B 3

C 5

D 7

E 8

WORKED SOLUTION

Multiplying A_n by $x + x^{-1} = 3$ gives

$$3A_n = A_{n+1} + A_{n-1},$$

so $A_{n+1} = 3A_n - A_{n-1}$. Also $A_0 = 2$ and $A_1 = 3$. Working modulo 10,

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6 \equiv 2, 3, 7, 8, 7, 3, 2.$$

The pair $(A_6, A_7) \equiv (2, 3)$ repeats the initial pair, so the remainders have period 6. Since 2025 leaves remainder 3 on division by 6,

$$A_{2025} \equiv A_3 \equiv 8 \pmod{10}.$$

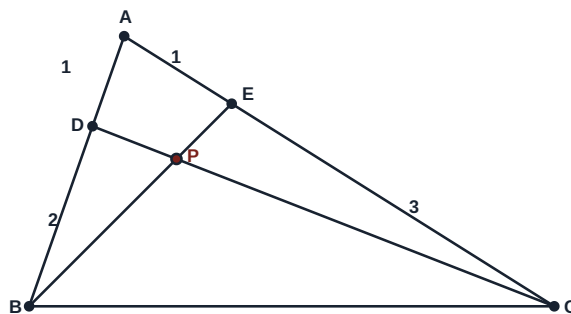
Therefore the correct option is E.

Trap check: Track a pair of consecutive remainders to justify the cycle; matching one term alone is not enough.

Question 27

In triangle ABC , point D lies on AB with $AD : DB = 1 : 2$, and point E lies on AC with $AE : EC = 1 : 3$. Lines DC and EB intersect at P .

What is $EP : PB$?



A 3 : 7

B 8 : 3

C 3 : 8

D 4 : 9

E 2 : 7

F 3 : 11

G 11 : 3

H 8 : 11

WORKED SOLUTION

Choose convenient coordinates $A = (0, 0)$, $B = (3, 0)$, $C = (0, 4)$. Then $D = (1, 0)$ and $E = (0, 1)$. Parameterise P on both cevians:

$$P = D + t(C - D) = (1 - t, 4t),$$

$$P = E + s(B - E) = (3s, 1 - s).$$

Solving $1 - t = 3s$ and $4t = 1 - s$ gives $s = 3/11$. Since s is the fraction of the way from E to B ,

$$EP : PB = s : (1 - s) = \frac{3}{11} : \frac{8}{11} = 3 : 8.$$

Therefore the correct option is **C**.

Trap check: B reverses the required ratio; F reports $EP : EB = 3 : 11$; G reverses that whole-segment ratio; H reports $PB : EB = 8 : 11$. A, D and E arise from using one of the side ratios directly or mismatching coefficients in the two line parameters.

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 2

27 hard, recall-informed Mathematics 2 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 2 answer key

Q1	B	Q2	D	Q3	A	Q4	C	Q5	A	Q6	E	Q7	D	Q8	A	Q9	B
Q10	B	Q11	E	Q12	D	Q13	C	Q14	A	Q15	C	Q16	C	Q17	A	Q18	D
Q19	C	Q20	C	Q21	D	Q22	E	Q23	B	Q24	E	Q25	B	Q26	E	Q27	B

Mathematics 2

27 hard, recall-informed Mathematics 2 questions with complete worked solutions.

M2-01

Geometric progressions

Answer **B**

Question 1

A geometric progression has positive terms, first term a and common ratio r . The sum of its first three terms is **14**, and the sum of its first six terms is **126**. What is $a + r^2$?

A 4**B** 6**C** 8**D** 10**E** 14

WORKED SOLUTION

Let the sum of the first three terms be $S_3 = 14$. The next three terms are obtained by multiplying the first three by r^3 , so

$$S_6 = S_3(1 + r^3).$$

Hence $126 = 14(1 + r^3)$, giving $r^3 = 8$. Positivity selects $r = 2$. Now

$$14 = a(1 + r + r^2) = 7a,$$

so $a = 2$. Therefore $a + r^2 = 2 + 4 = 6$.

Question 2

The polynomial $p(x) = x^3 + ax^2 + bx + 6$ is divisible by $x - 3$. It also leaves the same remainder when divided by $x - 1$ and by $x + 2$. What is $p(2)$?

A -11

B -9

C -5

D -4

E -7

F -2

WORKED SOLUTION

By the factor theorem, $p(3) = 0$, so

$$27 + 9a + 3b + 6 = 0 \implies 3a + b = -11.$$

Equal remainders give $p(1) = p(-2)$:

$$7 + a + b = -2 + 4a - 2b \implies a - b = 3.$$

Solving gives $a = -2$ and $b = -5$. Therefore

$$p(2) = 8 + 4(-2) + 2(-5) + 6 = -4.$$

Question 3

The table shows three values of a twice-differentiable function f .

x	0	1	3
$f(x)$	4	7	5

It is known that $f''(x) < 0$ for $0 < x < 3$. The trapezium rule is applied separately on $[0, 1]$ and $[1, 3]$. Which statement is correct?

A The estimate is $\frac{35}{2}$ and is an underestimate.B The estimate is $\frac{35}{2}$ and is an overestimate.C The estimate is $\frac{23}{2}$ and is an underestimate.D The estimate is $\frac{27}{2}$ and is an underestimate.

E The estimate is 23 and is an overestimate.

WORKED SOLUTION

The unequal strip widths must be retained:

$$T = \frac{1}{2}(4 + 7) + \frac{2}{2}(7 + 5) = \frac{11}{2} + 12 = \frac{35}{2}.$$

Since $f''(x) < 0$, the graph is concave down. Each chord lies below the graph, so the trapezium-rule estimate is an underestimate.

Question 4

For an arithmetic progression, the sum of the first **10** terms is **5**, and the sum of the first **20** terms is **110**. What is the sum of the positive terms among its first **30** terms?

A 285**B** 300**C** 325**D** 315**E** 330**WORKED SOLUTION**

For first term a and common difference d ,

$$S_{10} = 5(2a + 9d) = 5, \quad S_{20} = 10(2a + 19d) = 110.$$

Thus $2a + 9d = 1$ and $2a + 19d = 11$, so $d = 1$ and $a = -4$. The first 30 terms are therefore

$$-4, -3, -2, -1, 0, 1, 2, \dots, 25.$$

The positive terms are **1** to **25**, with sum

$$\frac{25 \cdot 26}{2} = 325.$$

Question 5

The curve $y = x^3 - 6x^2 + 9x + 4$ has two stationary points. The straight line joining them meets the x -axis at $(q, 0)$. What is q ?

A 5**B** 4**C** 6**D** 3**E** 2**WORKED SOLUTION**

Differentiating gives $y' = 3(x - 1)(x - 3)$, so the stationary points are $(1, 8)$ and $(3, 4)$. The line joining them has gradient

$$\frac{4 - 8}{3 - 1} = -2,$$

so its equation is $y - 8 = -2(x - 1)$, or $y = 10 - 2x$. Setting $y = 0$ gives $q = 5$.

Question 6Solve $\log_2(x^2 - 5x + 6) > 0$.

A $2 < x < 3$

B $x < 2$ or $x > 3$

C $\frac{5-\sqrt{5}}{2} < x < \frac{5+\sqrt{5}}{2}$

D $x < \frac{5-\sqrt{5}}{2}$ or $x > 3$

E $x < \frac{5-\sqrt{5}}{2}$ or $x > \frac{5+\sqrt{5}}{2}$

F $x \leq \frac{5-\sqrt{5}}{2}$ or $x \geq \frac{5+\sqrt{5}}{2}$

G $x < \frac{5-\sqrt{5}}{2}$

H $x > \frac{5+\sqrt{5}}{2}$

WORKED SOLUTION

Because the base is greater than 1, the inequality is equivalent to

$$x^2 - 5x + 6 > 1 \iff x^2 - 5x + 5 > 0.$$

The roots are $\frac{5 \pm \sqrt{5}}{2}$, and the quadratic is positive outside them. On this set the logarithm's argument is automatically greater than 1, so the domain condition is satisfied.**Question 7**An acute angle θ satisfies $\tan \theta + \sec \theta = 3$. What is $\sin 2\theta$?

A $\frac{4}{5}$

B $\frac{3}{5}$

C $\frac{16}{25}$

D $\frac{24}{25}$

E $\frac{25}{24}$

WORKED SOLUTIONUse $\sec^2 \theta - \tan^2 \theta = 1$:

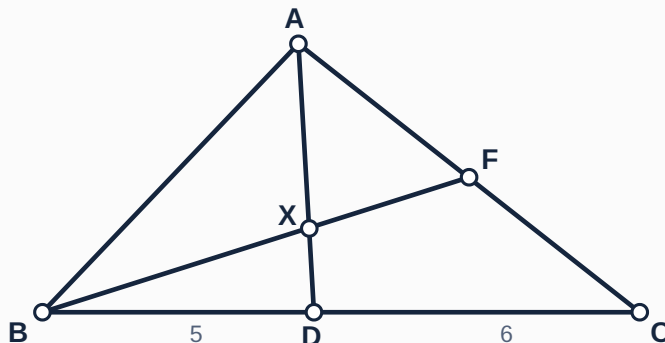
$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1.$$

Therefore $\sec \theta - \tan \theta = 1/3$. Solving the two linear equations gives $\sec \theta = 5/3$ and $\tan \theta = 4/3$. Since θ is acute, $\cos \theta = 3/5$ and $\sin \theta = 4/5$. Hence

$$\sin 2\theta = 2 \sin \theta \cos \theta = \frac{24}{25}.$$

Question 8

In triangle ABC , point D lies on BC with $BD : DC = 5 : 6$, and F is the midpoint of AC . The lines AD and BF meet at X . Find $AX : XD$.

**A** 11 : 5**B** 5 : 11**C** 11 : 6**D** 6 : 11**E** 16 : 5**WORKED SOLUTION**

Take A as the origin and write the position vectors of B, C as $\mathbf{b}, \mathbf{c}$. Then

$$\mathbf{d} = \frac{6\mathbf{b} + 5\mathbf{c}}{11}, \quad \mathbf{f} = \frac{\mathbf{c}}{2}.$$

A point on AD has vector $t\mathbf{d}$, while a point on BF has vector $(1-s)\mathbf{b} + \frac{s}{2}\mathbf{c}$. Equating coefficients gives

$$\frac{6t}{11} = 1 - s, \quad \frac{5t}{11} = \frac{s}{2}.$$

Hence $s = \frac{10t}{11}$ and $6t = 11 - 10t$, so $t = \frac{11}{16}$. Thus $AX : XD = t : (1-t) = 11 : 5$.

Question 9

A straight line with positive intercepts on the coordinate axes passes through $(2, 3)$. The triangle enclosed by the line and the positive coordinate axes has area **12**. What is the sum of the two intercepts?

A 8

B 10

C 12

D 11

E 9

F 24

WORKED SOLUTION

Let the intercepts be $a, b > 0$. The area condition gives $ab = 24$, and the intercept form of the line gives

$$\frac{2}{a} + \frac{3}{b} = 1.$$

Substituting $b = 24/a$ gives $2/a + a/8 = 1$, so $a^2 - 8a + 16 = 0$. Hence $a = 4$, $b = 6$, and $a + b = 10$.

Question 10

A positive geometric progression has sum to infinity **24**, and the sum of its first three terms is **21**. What is the sum of all terms whose positions are positive multiples of **3**?

A 8

B $\frac{24}{7}$ C $\frac{12}{7}$

D 3

E 6

WORKED SOLUTION

Since $S_3 = S_\infty(1 - r^3)$,

$$21 = 24(1 - r^3),$$

so $r^3 = \frac{1}{8}$ and, because the terms are positive, $r = \frac{1}{2}$. Then $a = 24(1 - r) = 12$. The terms in positions **3, 6, 9, ...** form

$$ar^2 + ar^5 + ar^8 + \dots,$$

a geometric series with first term $ar^2 = 3$ and ratio $r^3 = 1/8$. Its sum is

$$\frac{3}{1 - 1/8} = \frac{24}{7}.$$

Question 11Solve $2^x + 2^{-x} = \frac{5}{2}$.**A** $x = -1$ only**B** $x = 0, \pm 1$ **C** $x = 1$ only**D** $x = \pm 2$ **E** $x = \pm 1$ **WORKED SOLUTION**Let $u = 2^x > 0$. Then $u + u^{-1} = \frac{5}{2}$, so $2u^2 - 5u + 2 = 0$. Hence $u = 2$ or $u = \frac{1}{2}$, giving $x = 1$ or $x = -1$.**Question 12**

Let

$$A = \int_0^{1/2} (\sqrt{x} - x\sqrt{x})^2 dx, \quad B = \int_{1/2}^1 (\sqrt{x} - x\sqrt{x})^2 dx.$$

What is $A : B$?**A** 5 : 11**B** 16 : 5**C** 11 : 16**D** 11 : 5**E** 5 : 16**WORKED SOLUTION**

Expand before integrating:

$$(\sqrt{x} - x\sqrt{x})^2 = x - 2x^2 + x^3.$$

An antiderivative is

$$F(x) = \frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}.$$

Hence $A = F(1/2) - F(0) = 11/192$. Also $F(1) = 1/12 = 16/192$, so

$$B = F(1) - F(1/2) = \frac{5}{192}.$$

Therefore $A : B = 11 : 5$.

Question 13

The equation $x^2 - 4x + k = 0$ has two distinct real roots α and β . Given that $\alpha^4 + \beta^4 = 194$, find k .

A 31**B** -1**C** 1**D** 4**E** 32**WORKED SOLUTION**

Here $\alpha + \beta = 4$ and $\alpha\beta = k$. Thus

$$\alpha^2 + \beta^2 = 16 - 2k$$

and

$$\alpha^4 + \beta^4 = (16 - 2k)^2 - 2k^2 = 194.$$

This simplifies to $k^2 - 32k + 31 = 0$, so $k = 1$ or 31 . Two distinct real roots require the discriminant $16 - 4k > 0$, hence $k < 4$. Therefore $k = 1$.

Question 14

What is the coefficient of x^3 in $(1 + 2x)^5(1 - x)^3$?

A -11**B** -13**C** -9**D** 11**E** -15**WORKED SOLUTION**

The x^3 -coefficient is the sum of the four compatible products:

$$\binom{5}{3}2^3 - \binom{5}{2}2^2\binom{3}{1} + \binom{5}{1}2\binom{3}{2} - \binom{3}{3}.$$

This is $80 - 120 + 30 - 1 = -11$.

Question 15

For $k > 0$, the equation $|x^2 - 4x + 3| = k$ has exactly three distinct real solutions. What is the sum of those solutions?

A 4

B 2

C 6D $4 + \sqrt{2}$

E 8

WORKED SOLUTION

Write $x^2 - 4x + 3 = (x - 2)^2 - 1$. The branch $(x - 2)^2 - 1 = k$ has two roots for every $k > 0$. The other branch is

$$(x - 2)^2 - 1 = -k \iff (x - 2)^2 = 1 - k.$$

It contributes two roots if $0 < k < 1$, one root if $k = 1$, and none if $k > 1$. Exactly three roots therefore requires $k = 1$. They are $2 - \sqrt{2}, 2, 2 + \sqrt{2}$, with sum 6.

Question 16

The line $y = 2x + 1$ cuts the circle $x^2 + y^2 = 25$ in a chord. What is the area of the triangle whose vertices are the origin and the two endpoints of the chord?

A $\frac{\sqrt{31}}{5}$ B $\frac{4\sqrt{31}}{5}$ **C $\frac{2\sqrt{31}}{5}$** D $2\sqrt{31}$ E $\sqrt{31}$ **WORKED SOLUTION**

The perpendicular distance from the origin to $2x - y + 1 = 0$ is $d = 1/\sqrt{5}$. If h is half the chord length, then

$$h^2 + d^2 = 5^2 \implies h^2 = 25 - \frac{1}{5} = \frac{124}{5}.$$

The full chord length is $2h = 2\sqrt{124/5} = 4\sqrt{31/5}$. Taking this chord as the base and d as the perpendicular height, the triangle area is

$$\frac{1}{2}(2h)d = \sqrt{\frac{124}{5}} \cdot \frac{1}{\sqrt{5}} = \frac{2\sqrt{31}}{5}.$$

Question 17

For $f(x) = x^2 - 4x + 3$, evaluate

$$\frac{\int_0^4 |f(x)| dx}{\left| \int_0^4 f(x) dx \right|}.$$

A 3**B** 2**C** 1**D** 4**E** $\frac{1}{3}$ **WORKED SOLUTION**

The curve crosses the axis at $x = 1, 3$, and is below it between those values. With

$$F(x) = \frac{x^3}{3} - 2x^2 + 3x,$$

the three unsigned region areas are each $\frac{4}{3}$, so

$$\int_0^4 |f(x)| dx = 4.$$

The signed integral is $\frac{4}{3} - \frac{4}{3} + \frac{4}{3} = \frac{4}{3}$. The required ratio is therefore

$$\frac{4}{4/3} = 3.$$

Question 18

The curve $y = x^3 - 3x^2 + kx$ has two distinct stationary points. The gradient of the straight line joining them is -1 . Find k .

A $\frac{1}{2}$ **B** 1**C** 3**D** $\frac{3}{2}$ **E** $\frac{9}{2}$ **WORKED SOLUTION**

Let the stationary x -coordinates be r, s . Since

$$3x^2 - 6x + k = 0,$$

Vieta gives $r + s = 2$ and $rs = k/3$. For $f(x) = x^3 - 3x^2 + kx$, the secant gradient between r and s is

$$\frac{f(r) - f(s)}{r - s} = r^2 + rs + s^2 - 3(r + s) + k.$$

Now $r^2 + rs + s^2 = (r + s)^2 - rs = 4 - k/3$, so the gradient is $-2 + 2k/3$. Setting this equal to -1 gives $k = 3/2$. The derivative discriminant is then positive, so the stationary points are distinct.

Question 19

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 1$. If the real solutions of $f(g(x)) = g(f(x))$ are r and s , what is $f(r) + f(s)$?

A -2 **B** $\frac{1}{2}$ **C** 1**D** 2**E** 3**WORKED SOLUTION**

We have

$$f(g(x)) = (2x + 1)^2 - 1 = 4x^2 + 4x$$

and $g(f(x)) = 2(x^2 - 1) + 1 = 2x^2 - 1$. Hence

$$2x^2 + 4x + 1 = 0.$$

The discriminant is positive, so both roots are real. By Vieta, $r + s = -2$ and $rs = 1/2$. Therefore

$$f(r) + f(s) = r^2 + s^2 - 2 = (r + s)^2 - 2rs - 2 = 4 - 1 - 2 = 1.$$

Question 20

The solutions of $2\sin^2 x + \sin x - 1 = 0$ for $0 \leq x < 2\pi$ have sum

A 3π

B $\frac{11\pi}{4}$

C $\frac{5\pi}{2}$

D 2π

E $\frac{9\pi}{4}$

WORKED SOLUTION

Factorising gives $(2\sin x - 1)(\sin x + 1) = 0$. Thus

$$x = \frac{\pi}{6}, \quad \frac{5\pi}{6}, \quad \frac{3\pi}{2}.$$

Their sum is $\frac{5\pi}{2}$.

Question 21

Let $f(x) = |x^2 - 4x + 3|$. How many real solutions does $f(|x|) = 1$ have?

A 2

B 3

C 4

D 6

E 8

F 5

G 7

H 12

WORKED SOLUTION

Put $t = |x|$, so $t \geq 0$. We need

$$|t^2 - 4t + 3| = 1.$$

The positive branch gives $(t - 2)^2 = 2$, hence $t = 2 \pm \sqrt{2}$, both positive. The negative branch gives $(t - 2)^2 = 0$, hence $t = 2$. Each of these three positive values of t corresponds to two values $x = \pm t$. Therefore there are **6** real solutions.

Question 22

The line $y = kx + 1$ intersects the parabola $y = x^2 - 4x + 5$ at two points whose x -coordinates differ by 4. What is the sum of all possible values of k ?

A -4 **B** $-4\sqrt{2}$ **C** 8 **D** 4 **E** -8 **WORKED SOLUTION**

At an intersection,

$$x^2 - (k + 4)x + 4 = 0.$$

If its roots are r, s , then $(r - s)^2 = (r + s)^2 - 4rs$. Here $r + s = k + 4$ and $rs = 4$, so

$$16 = (k + 4)^2 - 16.$$

Hence $(k + 4)^2 = 32$, giving $k = -4 \pm 4\sqrt{2}$. Their sum is -8 .

Question 23

Two distinct tangents drawn from the origin touch the parabola $y = x^2 - 4x + 5$. What is the product of the x -coordinates of the two points of contact?

A 0 **B** -5 **C** -8 **D** 5 **E** -4 **WORKED SOLUTION**

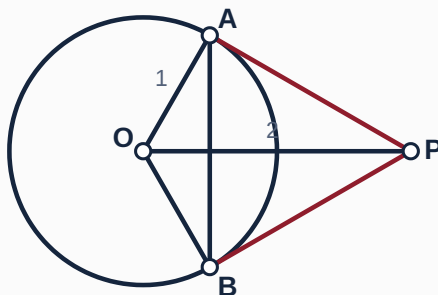
At $x = a$, the tangent has gradient $2a - 4$. Because it passes through the origin, its gradient is also $f(a)/a$. Therefore

$$a(2a - 4) = a^2 - 4a + 5 \implies a^2 = 5.$$

The contact coordinates have x -values $\sqrt{5}$ and $-\sqrt{5}$, whose product is -5 .

Question 24

A circle has centre O and radius 1 . A point P satisfies $OP = 2$, and the tangents from P touch the circle at A and B . What is the area of the smaller segment bounded by chord AB and the minor arc AB ?



A $\frac{\pi}{6} - \frac{\sqrt{3}}{8}$

B $\frac{\pi}{3} + \frac{\sqrt{3}}{4}$

C $\frac{2\pi}{3} - \frac{\sqrt{3}}{4}$

D $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$

E $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

F $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

G $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

H $\frac{\pi}{3} - \frac{\sqrt{3}}{8}$

WORKED SOLUTION

Since a radius is perpendicular to a tangent, triangle OAP is right-angled at A . Thus

$$\cos \angle AOP = \frac{OA}{OP} = \frac{1}{2},$$

so $\angle AOP = \pi/3$. By symmetry, $\angle AOB = 2\pi/3$. The sector area is $\frac{1}{2}(1)^2(2\pi/3) = \pi/3$, while

$$\text{area}(\triangle AOB) = \frac{1}{2} \sin(2\pi/3) = \frac{\sqrt{3}}{4}.$$

The required segment area is $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$.

Question 25

In triangle ABC , side a is opposite angle A , and side b is opposite angle B . Given $a = 2$, $b = 2\sqrt{3}$, and $A = 30^\circ$, two non-congruent triangles are possible. What is the sum of their areas?

A $\sqrt{3}$ **B** $3\sqrt{3}$ **C** $2\sqrt{3}$ **D** $4\sqrt{3}$ **E** 6**WORKED SOLUTION**

By the sine rule,

$$\sin B = \frac{b \sin A}{a} = \frac{2\sqrt{3}(1/2)}{2} = \frac{\sqrt{3}}{2}.$$

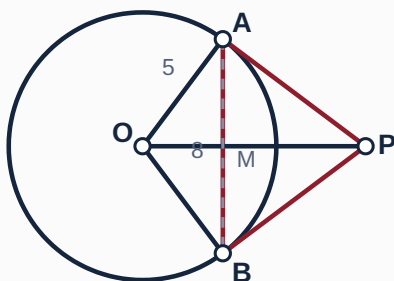
Hence $B = 60^\circ$ or 120° , giving $C = 90^\circ$ or 30° . Since the included angle between sides a, b is C , the two areas are

$$\frac{1}{2}ab \sin 90^\circ = 2\sqrt{3}, \quad \frac{1}{2}ab \sin 30^\circ = \sqrt{3}.$$

Their sum is $3\sqrt{3}$.

Question 26

A circle has centre O and radius 5 . A chord AB has length 8 . The tangents at A and B meet at P . Find the area of quadrilateral $OAPB$.



A $\frac{50}{3}$

B $\frac{80}{3}$

C 25

D $\frac{200}{3}$

E $\frac{100}{3}$

WORKED SOLUTION

Let M be the midpoint of AB . Then $AM = 4$, and $OM = \sqrt{5^2 - 4^2} = 3$. Symmetry puts O, M, P on one line. Triangles OMA and OAP are similar: both are right-angled and share the angle at O . Therefore

$$\frac{OP}{OA} = \frac{OA}{OM} \implies OP = \frac{25}{3}.$$

Hence $AP = \sqrt{OP^2 - OA^2} = 20/3$. Quadrilateral $OAPB$ is two congruent right triangles, so its area is

$$2 \left(\frac{1}{2} \cdot 5 \cdot \frac{20}{3} \right) = \frac{100}{3}.$$

Question 27

A curve satisfies $\frac{dy}{dx} = 3x(x - 2)$. It is tangent to the x -axis at its positive stationary point. What is the finite area enclosed by the curve and the x -axis?

A $\frac{9}{2}$

B $\frac{27}{4}$

C $\frac{11}{4}$

D 4

E 8

WORKED SOLUTION

Integrating gives $y = x^3 - 3x^2 + C$. The positive stationary point is at $x = 2$, and tangency to the x -axis means $y(2) = 0$, so $C = 4$. Thus

$$y = x^3 - 3x^2 + 4 = (x - 2)^2(x + 1).$$

The curve meets the axis at $x = -1$ and touches it at $x = 2$; it is non-negative between them. Hence the enclosed area is

$$\int_{-1}^2 (x^3 - 3x^2 + 4) dx = \left[\frac{x^4}{4} - x^3 + 4x \right]_{-1}^2 = 4 - \left(-\frac{11}{4} \right) = \frac{27}{4}.$$

ANSWER KEY

ESAT 2024-25 Recall Mock - Chemistry

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Chemistry answer key

Q1	C	Q2	A	Q3	E	Q4	B	Q5	D	Q6	B	Q7	A	Q8	D	Q9	C
Q10	E	Q11	C	Q12	A	Q13	E	Q14	B	Q15	D	Q16	A	Q17	C	Q18	E
Q19	B	Q20	D	Q21	E	Q22	C	Q23	A	Q24	D	Q25	B	Q26	C	Q27	A

Chemistry

27 hard, recall-informed Chemistry questions with complete worked solutions.

C-01

Isotopes and probability

Answer **C**

Question 1

Element X has two naturally occurring isotopes, ^{63}X and ^{65}X . Its relative atomic mass is 63.5. Atoms combine at random to form X_2 molecules. What percentage of X_2 molecules are expected to have relative molecular mass 128?

A 12.5%**B** 25.0%**C** 37.5%**D** 50.0%**E** 75.0%

WORKED SOLUTION

Let the fraction of ^{65}X be p . Then $63(1-p) + 65p = 63.5$, so $p = 0.25$ and the ^{63}X fraction is 0.75. Mass 128 requires one atom of each isotope, in either order: $2(0.75)(0.25) = 0.375 = 37.5\%$.

Therefore the correct option is **C**.

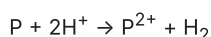
C-02

Ions, formulae and limiting amounts

Answer **A**

Question 2

Atoms P and Q have electron configurations P: 2,8,2 and Q: 2,7. In a two-stage process, 0.240 mol of P atoms and 0.360 mol of Q atoms first react as completely as possible. The ionic solid is removed, and any unreacted P is then added to excess acid:



Which option correctly gives the ionic solid formed in the first stage and the amount of hydrogen formed in the second?

A 0.180 mol of PQ_2 ; 0.060 mol H_2 **B** 0.240 mol of PQ_2 ; no H_2 **C** 0.120 mol of P_2Q ; 0.120 mol H_2 **D** 0.180 mol of covalent PQ_2 ; 0.060 mol H_2 **E** 0.360 mol of PQ_2 ; 0.120 mol H_2

WORKED SOLUTION

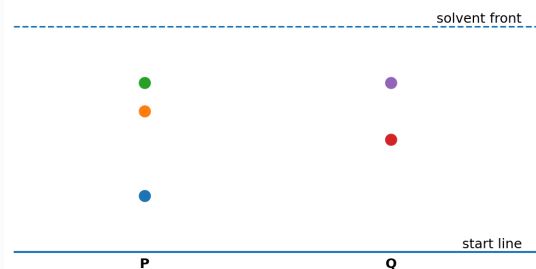
P forms P^{2+} and Q forms Q^- , so the ionic formula is PQ_2 . The 0.360 mol of Q atoms can form $0.360/2 = 0.180$ mol PQ_2 , using 0.180 mol P. This leaves $0.240 - 0.180 = 0.060$ mol P. The acid equation has a 1:1 ratio between P and H_2 , so the second stage forms 0.060 mol H_2 .

Therefore the correct option is **A**.

Distractor check: B ignores the 1:2 formula ratio and the leftover P; C reverses the ionic formula and doubles the gas; D has the right amounts but the wrong bonding type; E equates the amount of Q atoms with the amount of compound.

Question 3

The chromatograms P and Q were run with the same solvent and stationary phase. The solvent front travelled 8.0 cm in each case. In P, spot distances from the start are 2.0, 5.0 and 6.0 cm. In Q, they are 4.0 and 6.0 cm. Reference substances have R_f values W=0.25, X=0.50, Y=0.625 and Z=0.75. Which statement is correct?



- A** P contains W, X and Y
- B** P and Q both contain X
- C** Only Q contains Z
- D** Q contains W and Y
- E** P contains W, Y and Z, while Q contains X and Z

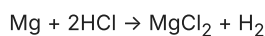
WORKED SOLUTION

For P, R_f values are 0.25, 0.625 and 0.75, so P contains W, Y and Z. For Q they are 0.50 and 0.75, so Q contains X and Z.

Therefore the correct option is **E**.

Question 4

2.40 g of magnesium is added to 100 cm³ of 1.00 mol dm⁻³ hydrochloric acid. When gas production stops, the unreacted magnesium is separated without loss and added to 100 cm³ of 0.500 mol dm⁻³ hydrochloric acid.



Take $A_r(\text{Mg})=24$ and molar gas volume at rtp as 24 dm³ mol⁻¹. What total volume of hydrogen is produced across the two stages?

A 0.60 dm³

B 1.80 dm³

C 2.40 dm³

D 1.20 dm³

E 3.60 dm³

WORKED SOLUTION

Initially there are $2.40/24=0.100$ mol Mg and 0.100 mol HCl. The first acid portion therefore forms 0.050 mol H₂ and consumes 0.050 mol Mg, leaving 0.050 mol Mg. The second acid portion contains $0.100 \times 0.500 = 0.050$ mol HCl, so it forms a further 0.025 mol H₂. The total is 0.075 mol, occupying $0.075 \times 24 = 1.80$ dm³ at rtp.

Therefore the correct option is **B**.

Distractor check: A counts only the second stage; C assumes all the original magnesium reacts; D counts only the first stage; E treats the HCl:H₂ ratio as 1:1 in both stages.

Question 5

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, the forward reaction is exothermic. Which single change increases both the equilibrium yield of ammonia and the initial forward reaction rate?

A decrease temperature

B increase temperature

C add a catalyst

D decrease the container volume

E remove the catalyst

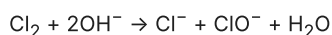
WORKED SOLUTION

Decreasing volume increases pressure, favouring the side with fewer gas molecules (the products), and also increases collision frequency. A catalyst increases rate but not equilibrium yield; lowering temperature improves yield but slows the reaction.

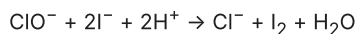
Therefore the correct option is **D**.

Question 6

In stage 1, 0.030 mol Cl_2 reacts completely with excess hydroxide ions:



The entire solution is then acidified and mixed with excess iodide ions. In stage 2, all the ClO^- reacts:



Which option gives the final amounts of Cl^- and I_2 , and the role of Cl_2 in stage 1?

- A** 0.030 mol Cl^- ; 0.030 mol I_2 ; both oxidising and reducing agent
- B** 0.060 mol Cl^- ; 0.030 mol I_2 ; both oxidising and reducing agent
- C** 0.060 mol Cl^- ; 0.060 mol I_2 ; oxidising agent only
- D** 0.030 mol Cl^- ; 0.060 mol I_2 ; reducing agent only
- E** 0.090 mol Cl^- ; 0.030 mol I_2 ; neither

WORKED SOLUTION

Stage 1 forms 0.030 mol Cl^- and 0.030 mol ClO^- . Stage 2 converts the 0.030 mol ClO^- into a further 0.030 mol Cl^- and forms 0.030 mol I_2 . The final chloride amount is therefore 0.060 mol. In stage 1, chlorine changes from oxidation state 0 to -1 in Cl^- and to +1 in ClO^- , so Cl_2 acts as both oxidising and reducing agent.

Therefore the correct option is **B**.

Distractor check: A omits the chloride made in stage 2; C doubles the iodine and ignores disproportionation; D reverses both the second-stage ratio and the redox role; E counts three chloride-producing routes although only two occur.

Question 7

Two pure solids are tested.

solid	melting point	electrical conduction	mechanical behaviour	water solubility
R	high	solid and molten	bends without shattering	insoluble
S	high	molten only	shatters when struck	soluble

Which explanation best fits all the evidence?

A

R is metallic, with delocalised electrons; S is ionic, with ions mobile only when molten.

B

R is ionic; S is metallic.

C

R is graphite; S is ionic.

D

R is metallic; S is giant covalent.

E

R is simple molecular; S is ionic.

WORKED SOLUTION

R's high melting point, conduction in both states and ability to bend are consistent with a metal: delocalised electrons carry charge and layers of positive ions can slide while metallic bonding persists. S is consistent with an ionic solid: its ions are fixed in the solid but mobile in the melt; brittleness and water solubility provide supporting evidence.

Therefore the correct option is **A**.

Distractor check: B reverses the conductivity and mechanical patterns. C fits several observations but graphite is brittle rather than malleable. D fits R completely and some of S, but a typical giant covalent solid neither dissolves in water nor conducts when molten. E fits S but conflicts with R's high melting point and conduction.

Question 8

A halogen X_2 does not displace chloride ions from solution, but does displace bromide and iodide ions. Which statement is correct?

A

X is bromine; X_2 is oxidised when it reacts with bromide ions.

B

X is chlorine; bromine can oxidise X^- ions.

C

X is fluorine; X_2 is reduced only when it reacts with iodide ions.

D

X is chlorine; X_2 is reduced when it reacts with bromide or iodide ions.

E

X is bromine; iodine can oxidise X^- ions.

WORKED SOLUTION

The displacement pattern identifies X as chlorine: chlorine does not displace chloride ions, but it is more reactive than bromine and iodine and so oxidises Br^- and I^- . In each displacement reaction, Cl_2 gains electrons to form Cl^- , so X_2 is reduced. Bromine cannot oxidise chloride ions because bromine is less reactive than chlorine.

Therefore the correct option is **D**.

Question 9

In each experiment, the same amount of calcium carbonate reacts with the same volume of excess hydrochloric acid.

experiment	acid concentration / mol dm ⁻³	temperature / °C	carbonate form	time to collect 40 cm ³ CO ₂ / s	final CO ₂ volume / cm ³
X	1.0	20	chips	80	48
Y	2.0	20	chips	45	48
Z	1.0	30	powder	20	48

Which conclusion is justified by the data?

- A** Y produces a greater final amount of carbon dioxide than X.
- B** Z proves that raising temperature alone makes the reaction faster.
- C** X and Y support the conclusion that higher acid concentration increases rate but not the final amount when carbonate is limiting.
- D** Equal final volumes show that all three reactions have the same rate.
- E** Z must contain the least calcium carbonate because it reaches 40 cm³ first.

WORKED SOLUTION

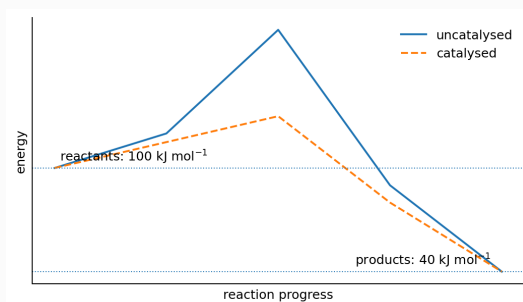
X and Y differ only in acid concentration. Y reaches 40 cm³ sooner, so it has a higher rate, while both reach 48 cm³ because the same amount of limiting calcium carbonate reacts. X and Z are not a controlled comparison for temperature because particle size also changes.

Therefore the correct option is **C**.

Distractor check: A contradicts the final volumes; B ignores the simultaneous surface-area change; D confuses amount with rate; E confuses reaching a volume sooner with producing less gas overall.

Question 10

The energy profile shows reactants at 100 kJ mol^{-1} , products at 40 kJ mol^{-1} , and a catalysed peak at 130 kJ mol^{-1} . What are ΔH for the forward reaction and the activation energy of the catalysed reverse reaction?

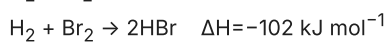
**A** -60 and 30 kJ mol^{-1} **B** $+60$ and 30 kJ mol^{-1} **C** $+60$ and 90 kJ mol^{-1} **D** -30 and 60 kJ mol^{-1} **E** -60 and 90 kJ mol^{-1} **WORKED SOLUTION**

$\Delta H = 40 - 100 = -60 \text{ kJ mol}^{-1}$. For the catalysed reverse reaction, $E_a = 130 - 40 = 90 \text{ kJ mol}^{-1}$.

Therefore the correct option is **E**.

Question 11

The following gaseous reactions have the stated enthalpy changes:



The average bond energies are $\text{H-H}=436$, $\text{Cl-Cl}=242$ and $\text{Br-Br}=194 \text{ kJ mol}^{-1}$. By how much is the average H-Cl bond energy greater than the average H-Br bond energy?

A 32.5 kJ mol^{-1}

B 41 kJ mol^{-1}

C 65 kJ mol^{-1}

D 82 kJ mol^{-1}

E 130 kJ mol^{-1}

WORKED SOLUTION

For HCl , $-184 = (436 + 242) - 2E(\text{H-Cl})$, so $E(\text{H-Cl}) = 862/2 = 431 \text{ kJ mol}^{-1}$. For HBr , $-102 = (436 + 194) - 2E(\text{H-Br})$, so $E(\text{H-Br}) = 732/2 = 366 \text{ kJ mol}^{-1}$. The difference is $431 - 366 = 65 \text{ kJ mol}^{-1}$.

Therefore the correct option is **C**.

Distractor check: A halves the final difference again; B uses only half the difference between the reaction enthalpies; D reports the reaction-enthalpy difference; E forgets that two H-X bonds form in each reaction.

Question 12

A strong acid has pH 2.00. A 10.0 cm^3 portion is diluted with water to 100.0 cm^3 . Then 20.0 cm^3 of this diluted acid is mixed with 180 cm^3 of $1.00\times 10^{-4}\text{ mol dm}^{-3}$ sodium hydroxide.

Assume complete dissociation and neutralisation, additive volumes and negligible ionisation of water. What is the final pH?

A 5.00**B** 3.00**C** 4.00**D** 6.00**E** 7.00**WORKED SOLUTION**

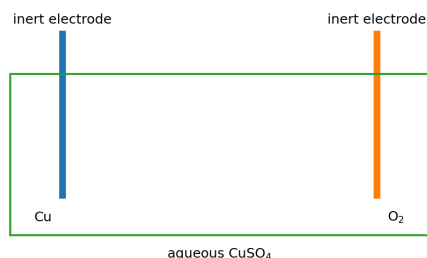
The original acid has $[\text{H}^+] = 1.00\times 10^{-2}\text{ mol dm}^{-3}$. Tenfold dilution makes this $1.00\times 10^{-3}\text{ mol dm}^{-3}$. The 20.0 cm^3 portion contains $2.00\times 10^{-5}\text{ mol H}^+$, while the alkali contains $0.180\times 1.00\times 10^{-4} = 1.80\times 10^{-5}\text{ mol OH}^-$. After neutralisation, $2.00\times 10^{-6}\text{ mol H}^+$ remains in 0.200 dm^3 , giving $[\text{H}^+] = 1.00\times 10^{-5}\text{ mol dm}^{-3}$ and pH 5.00.

Therefore the correct option is **A**.

Distractor check: B stops after the first dilution; C treats the mixing as dilution only and ignores hydroxide; D uses 2.00 dm^3 instead of 0.200 dm^3 as the final volume; E assumes exact neutralisation.

Question 13

Aqueous copper(II) sulfate is electrolysed with inert electrodes. Assume Cu^{2+} ions remain in excess throughout and that no other electrode reaction occurs. If 0.025 mol of oxygen is formed at the anode, what mass of copper is deposited at the cathode? Take $A_r(\text{Cu})=63.5$.

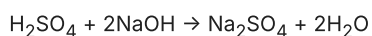
**A** 1.59 g**B** 6.35 g**C** 12.7 g**D** 25.4 g**E** 3.18 g**WORKED SOLUTION**

Forming 1 mol O_2 corresponds to 4 mol electrons, so 0.025 mol O_2 corresponds to 0.100 mol e^- . Cu^{2+} needs 2 e^- per Cu, so 0.050 mol Cu forms: $0.050 \times 63.5 = 3.175 \text{ g} \approx 3.18 \text{ g}$.

Therefore the correct option is **E**.

Question 14

25.0 cm³ of 0.200 mol dm⁻³ sulfuric acid is mixed with 40.0 cm³ of 0.250 mol dm⁻³ sodium hydroxide. Assume complete neutralisation and additive volumes.



A 13.0 cm³ aliquot of the resulting 65.0 cm³ mixture is removed. Then 5.0 cm³ of 0.200 mol dm⁻³ sodium hydroxide is added to what remains. Which statement describes the final mixture?

- A** It is neutral and contains 0.0040 mol of Na₂SO₄.
- B** It is alkaline and contains 0.0040 mol of Na₂SO₄.
- C** It is alkaline and contains 0.0050 mol of Na₂SO₄.
- D** It is neutral and contains 0.0050 mol of Na₂SO₄.
- E** It is acidic and contains 0.0010 mol of Na₂SO₄.

WORKED SOLUTION

Initially there are 0.00500 mol H₂SO₄ and 0.0100 mol NaOH, so they react exactly and form 0.00500 mol Na₂SO₄. Removing 13.0/65.0 = 1/5 of the homogeneous mixture removes 1/5 of the salt, leaving 0.00400 mol. The added sodium hydroxide is now in excess because no acid remains, so the final mixture is alkaline; it does not change the amount of Na₂SO₄.

Therefore the correct option is **B**.

Distractor check: A ignores the final sodium hydroxide; C ignores removal of the aliquot; D ignores both later operations; E reports the removed salt amount and reverses the acid–base conclusion.

Question 15

A 0.280 g sample of gaseous hydrocarbon X occupies 240 cm³ at rtp. X decolourises bromine water. Take the molar gas volume at rtp as 24 dm³ mol⁻¹.

Which option gives the formula of X and the formula of the organic product when X reacts with bromine?

A CH₄; CH₃Br

B C₂H₆; C₂H₆Br₂

C C₂H₄; C₂H₅Br

D C₂H₄; C₂H₄Br₂

E C₃H₆; C₃H₆Br₂

WORKED SOLUTION

The gas amount is $0.240/24 = 0.010$ mol, so $M_r(X) = 0.280/0.010 = 28$. The hydrocarbon with M_r 28 is C₂H₄. Its decolourisation of bromine water is consistent with a C=C bond, and bromine adds across that bond to form C₂H₄Br₂.

Therefore the correct option is **D**.

Distractor check: C treats addition as substitution; E uses the formula of the next alkene without reconciling it with the measured molar mass.

Question 16

A 10.0 g sample of a metal carbonate is heated, but the decomposition is stopped before completion:



The heating produces 1.92 dm³ CO₂ at rtp. The cooled solid is then treated with excess acid, producing a further 0.48 dm³ CO₂ at rtp. In the acid, each mole of undecomposed MCO₃ produces one mole of CO₂, while MO produces no gas. Take molar gas volume = 24 dm³ mol⁻¹, $A_r(\text{C}) = 12$ and $A_r(\text{O}) = 16$. What is $A_r(\text{M})$?

A 40

B 65

C 100

D 440

E 160

WORKED SOLUTION

Each mole of MCO₃ produces one mole of CO₂, whether it decomposes during heating or reacts later with acid. The total gas volume is $1.92 + 0.48 = 2.40$ dm³, corresponding to $2.40/24 = 0.100$ mol MCO₃. Therefore $M_r(\text{MCO}_3) = 10.0/0.100 = 100$, so $A_r(\text{M}) = 100 - (12 + 3 \times 16) = 40$.

Therefore the correct option is **A**.

Distractor check: B uses only the gas made during heating; C reports the relative formula mass of MCO₃; D uses only the undecomposed amount; E adds the carbonate-group mass instead of subtracting it.

Question 17

Two straight-chain compounds have condensed structural formulae $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}=\text{CHCH}_3$.

Which description is correct?

- A** They are the same compound written from opposite ends.
- B** They have different molecular formulae but belong to the same homologous series.
- C** They are structural isomers and both have molecular formula C_4H_8 .
- D** They have the same molecular formula but different functional groups.
- E** They are chain isomers and one is saturated.

WORKED SOLUTION

Both structures contain four carbon atoms, eight hydrogen atoms and one $\text{C}=\text{C}$ bond, so both have molecular formula C_4H_8 . The double bond is in a different position, so the displayed connectivity differs: they are structural (positional) isomers.

Therefore the correct option is **C**.

Distractor check: A confuses reversal of a formula with movement of the double bond; D overlooks that both compounds are alkenes.

Question 18

Metal M is below carbon in the reactivity series but above hydrogen. Consider the statements.

1. An oxide of M can be reduced by carbon under suitable conditions.
 2. M reacts with dilute hydrochloric acid to release hydrogen.
 3. During electrolysis of an aqueous salt of M using inert electrodes, M is deposited at the cathode in preference to hydrogen.
- Which statements are correct?

- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3
- E** 1 and 2 only

WORKED SOLUTION

Because M is below carbon, carbon can reduce its oxide under suitable conditions, so statement 1 is correct. Because M is above hydrogen, it can displace hydrogen from dilute acids, so statement 2 is correct. But for an aqueous electrolyte containing ions of a metal above hydrogen, hydrogen is discharged at the cathode in preference to the metal, so statement 3 is false.

Therefore the correct option is **E**.

Question 19

An aqueous salt gives a green precipitate with sodium hydroxide solution. A separate sample is acidified with dilute hydrochloric acid and then gives a white precipitate with barium chloride solution. Which inference is best?

- A** The salt contains Fe^{2+} and Cl^- ; acid is added to make the green precipitate form.
- B** The salt contains Fe^{2+} and SO_4^{2-} ; acidification prevents carbonate ions giving a misleading barium precipitate.
- C** The salt contains Cu^{2+} and SO_4^{2-} ; acid changes a blue precipitate to green.
- D** The salt contains Mg^{2+} and SO_4^{2-} ; barium chloride identifies the cation.
- E** The salt contains Fe^{2+} and NO_3^- ; hydrochloric acid supplies the sulfate ions.

WORKED SOLUTION

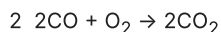
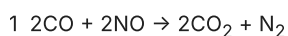
The green hydroxide precipitate indicates Fe^{2+} . A white precipitate after acidification and addition of barium chloride indicates sulfate ions. Acidification removes carbonate ions as carbon dioxide, preventing insoluble barium carbonate from causing a false positive.

Therefore the correct option is **B**.

Distractor check: A assigns the wrong anion and purpose to the acid; C and D mismatch the hydroxide colours; E wrongly claims hydrochloric acid contains sulfate.

Question 20

A 160 cm^3 exhaust pulse contains 90 cm^3 CO, 60 cm^3 NO and 10 cm^3 O₂. In a model catalytic converter, reaction 1 goes to completion before reaction 2 occurs:



All gas volumes are measured under the same conditions. Which option gives the final total gas volume and identifies the oxidising agent in each reaction?

A 90 cm^3 ; reaction 1: CO; reaction 2: CO

B 110 cm^3 ; reaction 1: NO; reaction 2: CO

C 120 cm^3 ; reaction 1: CO; reaction 2: NO

D 120 cm^3 ; reaction 1: NO; reaction 2: O₂

E 150 cm^3 ; reaction 1: O₂; reaction 2: O₂

WORKED SOLUTION

Reaction 1 consumes 60 cm^3 CO and all 60 cm^3 NO, forming 60 cm^3 CO₂ and 30 cm^3 N₂; 30 cm^3 CO remains. Reaction 2 uses all 10 cm^3 O₂ and 20 cm^3 of that CO, forming another 20 cm^3 CO₂. The final gases are 10 cm^3 CO, 80 cm^3 CO₂ and 30 cm^3 N₂, totalling 120 cm^3 . NO is reduced in reaction 1 and O₂ is reduced in reaction 2, so each is the oxidising agent in its reaction.

Therefore the correct option is **D**.

Distractor check: A treats the reducing agent as the oxidising agent in both stages; B identifies the first agent but loses a product volume and misclassifies CO in reaction 2; C finds the volume but reverses the redox roles; E ignores the first-stage reduction of NO and both gas contractions.

Question 21

A salt has solubility 80 g per 100 g water at 80°C and 20 g per 100 g water at 20°C. A 270 g saturated solution at 80°C is kept at 80°C while 30 g of water evaporates; any crystals formed remain in the vessel. The contents are then cooled to 20°C with no further water loss. What total mass of salt is present as crystals at 20°C?

A 24 g**B** 72 g**C** 90 g**D** 120 g**E** 96 g**WORKED SOLUTION**

At 80°C, 180 g saturated solution contains 100 g water and 80 g salt, so 270 g contains 150 g water and 120 g salt. Evaporation leaves 120 g water. At 80°C this can keep 96 g salt dissolved, so 24 g has already crystallised. At 20°C, 120 g water can hold only 24 g salt, so the total crystal mass is $120 - 24 = 96$ g.

Therefore the correct option is **E**.

Distractor check: A gives either the salt crystallised during evaporation or the salt left dissolved at 20°C; B counts only the additional crystallisation on cooling; C ignores the water loss; D assumes all salt crystallises.

Question 22

At the same temperature and pressure, 30 cm³ of gaseous A reacts exactly with 45 cm³ of gaseous B to form 30 cm³ of gaseous C only. Which equation has the simplest whole-number coefficients consistent with the data?

A $A + 1.5B \rightarrow C$ **B** $2A + 3B \rightarrow 3C$ **C** $2A + 3B \rightarrow 2C$ **D** $3A + 2B \rightarrow 3C$ **E** $4A + 6B \rightarrow 4C$ **WORKED SOLUTION**

At the same temperature and pressure, gas volumes are proportional to amounts in moles. The ratio 30:45:30 simplifies to 2:3:2. Therefore the simplest whole-number equation is $2A + 3B \rightarrow 2C$. Option E has the same ratio but its coefficients are not in simplest form.

Therefore the correct option is **C**.

Question 23

A student burns 0.010 mol of a fuel. The combustion raises the temperature of 200 g of water by 12.5°C. The specific heat capacity of water is 4.2 J g⁻¹ °C⁻¹.

Assume that 70% of the energy released by the fuel is transferred to the water. What is the molar energy release of the fuel?

A 1500 kJ mol⁻¹**B** 1050 kJ mol⁻¹**C** 735 kJ mol⁻¹**D** 150 kJ mol⁻¹**E** 15 kJ mol⁻¹**WORKED SOLUTION**

The water gains $q = mc\Delta T = 200 \times 4.2 \times 12.5 = 10\,500$ J. This is 70% of the fuel's energy release, so the fuel releases $10\,500 / 0.70 = 15\,000$ J for 0.010 mol. Per mole this is $15\,000 / 0.010 = 1\,500\,000$ J mol⁻¹ = 1500 kJ mol⁻¹.

Therefore the correct option is **A**.

Distractor check: B ignores the 70% transfer; C applies the efficiency in the wrong direction; D and E lose powers of ten during the per-mole or joule-to-kilojoule conversion.

Question 24

On complete combustion, assume that all sulfur in a fuel is converted to sulfur dioxide.

fuel	mass of sulfur burned	energy released
P	0.80 g	40 MJ
Q	0.30 g	10 MJ

Take $A_r(\text{S})=32$ and $A_r(\text{O})=16$. Which row gives the sulfur dioxide emitted per MJ and identifies the lower-emission fuel?

A P: 0.020 g MJ^{-1} ; Q: 0.030 g MJ^{-1} ; P is lower

B P: 0.040 g MJ^{-1} ; Q: 0.060 g MJ^{-1} ; Q is lower

C P: 0.080 g MJ^{-1} ; Q: 0.120 g MJ^{-1} ; P is lower

D P: 0.040 g MJ^{-1} ; Q: 0.060 g MJ^{-1} ; P is lower

E P: 0.040 g MJ^{-1} ; Q: 0.030 g MJ^{-1} ; Q is lower

WORKED SOLUTION

$M_r(\text{SO}_2)=64$, so each gram of sulfur forms $64/32=2 \text{ g}$ of SO_2 . Fuel P therefore emits $1.60/40=0.040 \text{ g MJ}^{-1}$; fuel Q emits $0.60/10=0.060 \text{ g MJ}^{-1}$. P has the lower emission per unit energy.

Therefore the correct option is **D**.

Distractor check: A compares sulfur rather than sulfur dioxide; B calculates both rates but reverses the comparison; C applies the mass factor twice.

Question 25

A mixture contains water, dissolved sodium chloride and insoluble sand. Which sequence can produce both pure water and dry sodium chloride as separate products?

- A** Filter the mixture, then evaporate the filtrate to dryness while allowing the water vapour to escape.
- B** Filter the mixture. Distil the filtrate and collect the water, stopping before the residue becomes dry. Evaporate the remaining concentrated solution and dry the sodium chloride crystals.
- C** Distil the original mixture and then dry its residue, without first separating the sand.
- D** Filter the mixture, crystallise some sodium chloride, and use the remaining mother liquor as the pure-water product.
- E** Centrifuge the mixture and decant the liquid as pure water, then dry the sediment as sodium chloride.

WORKED SOLUTION

Filtration removes the sand. Simple distillation produces pure water as the distillate. The remaining concentrated sodium chloride solution is then evaporated and the salt is dried.

Therefore the correct option is **B**.

Distractor check: A recovers salt but not water; C leaves sand mixed with the salt; D mistakes salt solution for pure water; E cannot remove dissolved sodium chloride by centrifugation.

Question 26

A 0.300 g compound contains only carbon, hydrogen and oxygen. Complete combustion produces 0.440 g CO_2 and 0.180 g H_2O . What is the empirical formula? Take A_r : C=12, H=1, O=16.

- A** CHO
- B** $\text{C}_2\text{H}_4\text{O}$
- C** CH_2O
- D** CH_4O
- E** $\text{C}_2\text{H}_2\text{O}$

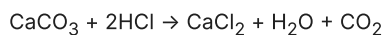
WORKED SOLUTION

$0.440 \text{ g CO}_2 = 0.010 \text{ mol CO}_2$, so the sample contained $0.010 \text{ mol C} = 0.120 \text{ g C}$. $0.180 \text{ g H}_2\text{O} = 0.010 \text{ mol H}_2\text{O}$, so it contained $0.020 \text{ mol H} = 0.020 \text{ g H}$. Oxygen mass is $0.300 - 0.140 = 0.160 \text{ g} = 0.010 \text{ mol O}$. Ratio C:H:O = 1:2:1, so CH_2O .

Therefore the correct option is **C**.

Question 27

A 10.0 g sample of impure calcium carbonate reacts with excess hydrochloric acid. A calibration shows that the gas-collection apparatus captures 75.0% of the carbon dioxide produced; the syringe reading is 1.26 dm³ at rtp. Assume that calcium carbonate is the only substance in the sample that produces carbon dioxide.



Take $M_r(\text{CaCO}_3)=100$ and molar gas volume = 24 dm³ mol⁻¹. What is the percentage purity?

A 70.0%**B** 52.5%**C** 35.0%**D** 93.3%**E** 28.0%**WORKED SOLUTION**

The actual carbon dioxide volume is $1.26/0.750=1.68$ dm³. This is $1.68/24=0.0700$ mol CO₂, so the sample contained 0.0700 mol CaCO₃, with mass $0.0700 \times 100 = 7.00$ g. The percentage purity is $(7.00/10.0) \times 100 = 70.0\%$.

Therefore the correct option is **A**.

Distractor check: B treats the captured volume as the full yield; C incorrectly applies the coefficient 2 from HCl; D corrects for the 75% capture twice; E uses $A_r(\text{Ca})=40$ instead of $M_r(\text{CaCO}_3)=100$.